

CODEVEDX AI/ML Internship Report

Project 2: Student Performance Prediction System (Machine Learning)

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Organization: CODEVEDX

Acknowledgement

I would like to express my sincere gratitude to CODEVEDX for providing me with the opportunity to work on this Artificial Intelligence & Machine Learning internship project. This project enhanced my knowledge of Python programming, data analysis, data visualization, and Machine Learning concepts.

Abstract

The **Student Performance Prediction System** is a Machine Learning-based application that predicts students' final marks using factors such as attendance, study hours, assignment scores, and internal marks. The project uses the **Linear Regression** algorithm to analyze historical student performance data and estimate future academic performance. It also includes data preprocessing, visualization, and model evaluation.

Introduction

Student performance prediction helps educational institutions identify students who may need additional academic support. By analyzing factors such as attendance, study hours, assignments, and internal marks, Machine Learning models can estimate students' final examination performance.

This project demonstrates how Artificial Intelligence and Machine Learning techniques can be used in the education sector for predictive analysis and better decision-making.

Objectives

- Develop a Machine Learning-based prediction system.
- Analyze student performance data.
- Perform data preprocessing.
- Visualize data using charts.
- Train a Linear Regression model.
- Predict students' final marks.
- Evaluate model accuracy.
- Gain practical experience in AI and Machine Learning.

Problem Statement

Educational institutions often need to predict students' academic performance to provide timely guidance and support. Manual prediction is difficult and time-consuming.

The objective of this project is to build a Machine Learning model that predicts students' final marks based on attendance, study hours, assignment scores, and internal marks.

Software Requirements

- Windows 10 / Windows 11
- Python 3.x
- Visual Studio Code
- Pandas
- Matplotlib
- Scikit-learn

Hardware Requirements

- Intel Core i3 or above
- Minimum 4 GB RAM
- 500 MB Free Storage
- Keyboard & Mouse

Technologies Used

Technology	Purpose
Python	Programming Language
Pandas	Data Loading & Analysis
Matplotlib	Data Visualization
Scikit-learn	Machine Learning Model
Linear Regression Performance Prediction	
CSV File	Dataset Storage

Modules Used

Module	Purpose
pandas	Reading CSV dataset
matplotlib.pyplot	Creating charts
sklearn.model_selection	Train-Test Split
sklearn.linear_model	Linear Regression
sklearn.metrics	Accuracy Evaluation

System Workflow

Start

↓

Load Dataset

↓

Data Preprocessing

↓

Data Visualization

↓

Train Machine Learning Model

↓

Predict Student Performance

↓

Display Accuracy

↓

End

Algorithm

Step 1

Load the student dataset (**Project2.csv**).

Step 2

Display the dataset and check for missing values.

Step 3

Visualize student performance using charts.

Step 4

Split the dataset into training and testing data.

Step 5

Train the Linear Regression model.

Step 6

Evaluate the model using the R^2 Score.

Step 7

Accept user input for prediction.

Step 8

Predict the student's final marks.

Step 9

Display the predicted marks.

Machine Learning Model

Algorithm Used: Linear Regression

Input Features

- Attendance
- Study Hours
- Assignment Score
- Internal Marks

Output

- Predicted Final Marks

Dataset

Dataset Name: Project2.csv

Dataset Description

Column Name	Description
Student_ID	Unique Student ID
Attendance	Attendance Percentage
Study_Hours	Daily Study Hours
Assignment_Score	Assignment Marks
Internal_Marks	Internal Exam Marks

Column Name	Description
Final_Marks	Final Examination Marks

Sample Dataset

Student ID	Attendance	Study Hours	Assignment	Internal	Final
1	90	5	85	80	84
2	85	4	80	75	78
3	70	2	65	60	62
4	95	6	90	88	91
5	80	3	75	72	74

Features

- View Dataset
- Dataset Information
- Data Visualization
- Train Machine Learning Model
- Predict Student Performance
- Model Accuracy Evaluation
- Exception Handling

Output Screenshots

Insert the following screenshots:

1. View Dataset Output

```
Enter Choice: 1
```

	Student_ID	Attendance	Study_Hours	Assignment_Score	Internal_Marks	Final_Marks
0	1	90	5	85	80	84
1	2	85	4	80	75	78
2	3	70	2	65	60	62
3	4	95	6	90	88	91
4	5	80	3	75	72	74
5	6	88	5	84	82	85
6	7	76	4	72	70	71
7	8	92	6	89	90	92
8	9	65	2	60	58	59
9	10	98	7	95	94	96
10	11	82	4	78	76	77
11	12	87	5	83	81	83
12	13	73	3	68	66	67
13	14	91	6	90	89	90
14	15	78	4	74	73	74
15	16	84	5	81	80	81
16	17	69	2	63	61	62
17	18	94	6	91	90	91
18	19	89	5	86	84	85
19	20	96	7	94	93	95

2. Dataset Information

```
Enter Choice: 2

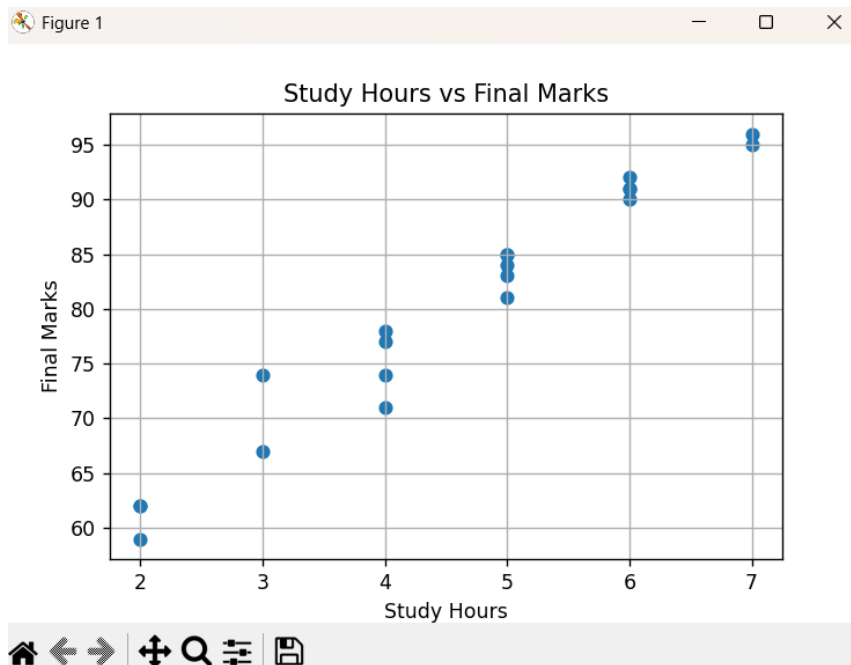
Dataset Information

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 6 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   Student_ID          20 non-null    int64
1   Attendance           20 non-null    int64
2   Study_Hours          20 non-null    int64
3   Assignment_Score     20 non-null    int64
4   Internal_Marks       20 non-null    int64
5   Final_Marks          20 non-null    int64
dtypes: int64(6)
memory usage: 1.1 KB
None

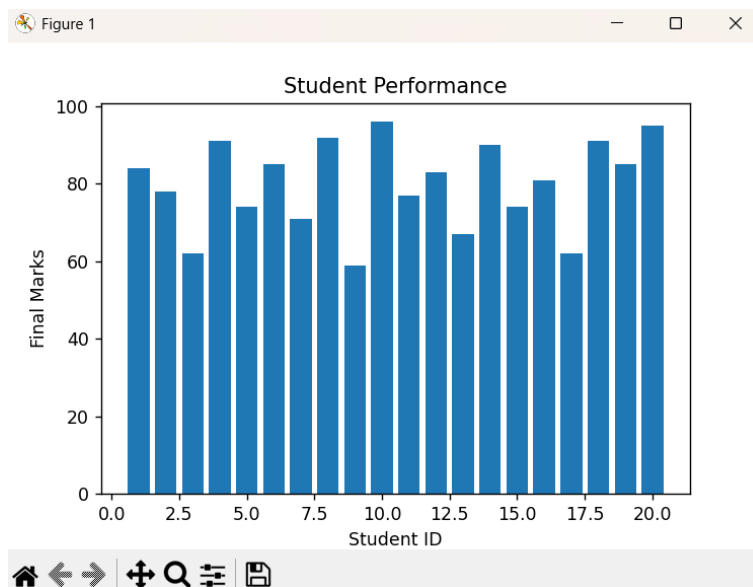
Missing Values

Student_ID      0
Attendance      0
Study_Hours     0
Assignment_Score 0
Internal_Marks  0
Final_Marks     0
dtype: int64
```

3. Scatter Plot



4. Bar Chart



5. Model Accuracy Output

```
STUDENT PERFORMANCE PREDICTION SYSTEM
1. View Dataset
2. Dataset Information
3. Data Visualization
4. Train Model
5. Predict Student Performance
6. Exit
Enter Choice: 4

Model Accuracy: 98.24 %
```

6. Student Performance Prediction Output

```
STUDENT PERFORMANCE PREDICTION SYSTEM
1. View Dataset
2. Dataset Information
3. Data Visualization
4. Train Model
5. Predict Student Performance
6. Exit
Enter Choice: 5

Model Accuracy: 98.24 %
Enter Attendance: 90
Enter Study Hours: 5
Enter Assignment Score: 85
Enter Internal Marks: 80

Predicted Final Marks: 83.21
```

Advantages

- Easy to use
- Fast prediction
- Accurate results
- Visual representation of data
- Beginner-friendly Machine Learning project

Limitations

- Uses a limited dataset.
- Prediction depends on data quality.
- Uses only Linear Regression.

Future Scope

- Add GUI using Tkinter.
- Deploy using Flask.
- Use larger datasets.
- Implement advanced Machine Learning algorithms such as Random Forest and XGBoost.
- Store records in a database.
- Generate performance reports automatically.

Conclusion

The **Student Performance Prediction System** was successfully developed using Python and the **Linear Regression** algorithm. The application predicts students' final marks based on academic factors such as attendance, study hours, assignment scores, and internal marks. This project enhanced my understanding of data analysis, visualization, Machine Learning, and predictive modeling.

References

1. Python Official Documentation
2. Pandas Official Documentation
3. Matplotlib Official Documentation
4. Scikit-learn Official Documentation
5. NumPy Official Documentation
6. Visual Studio Code Documentation